

Adenosine

(Adenocard, Adenoscan)

I. Classification – Antiarrhythmic**II. Action(s)**

- Slows AV node conduction time
- Interrupts AV Nodal re-entry pathways

III. Indication(s)

- Narrow Complex Tachycardia with Pulse
- Regular Wide Complex Tachycardia with Pulse (*avoid in Irregular Wide Complex*)

IV. Contraindications

- Known Hypersensitivity
- Irregular Wide Complex Tachycardia / Wolf-Parkinson-White (WPW)
- 2nd / 3rd Degree Heart Block without Pacemaker
- Sick Sinus Syndrome

V. Precautions

- Usually produces short period of asystole
- May induce significant arrhythmias including V-Tach / V-Fib
- May induce complete heart block
- Caution in history of COPD / Asthma (must have rescue airway available)

VI. Drug interactions

- Digoxin and Verapamil – May increase risk of V-Fib
- Carbamazepine – May increase degree of heart block
- Aspirin and Dipyridamole– May increase Adenosine effects

VII. Adult Dosing and Administration

- 12 mg rapid IV push, may repeat x1 in Narrow Complex Tachycardia with Pulse

VIII. Pediatric Dosing and Administration

- First Dose - 0.1 mg/kg (max dose 12 mg)
- Second Dose (Narrow Complex Tach with Pulse only) - 0.2 mg/kg (max dose 12 mg)
- Age 13 years and older, use Adult Dosing

IX. Onset

- Rapid (1-2 sec)

X. Duration

- Brief (10-20 sec)

XI. Possible Adverse Effects

- May induce significant arrhythmia including v-fib, v-tach, and bradycardia requiring resuscitation
- Expect a short period of asystole / bradycardia (10-15 sec)
- Commonly induces nausea, headache, chest pressure

XII. Other Notes

- Must have defibrillation / pacer pads attached to patient, ACLS medication, and rescue airway available prior to medication administration